

Understanding Linear Algebra

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1 Why Matrix Multiplication is defined as that form?

Let's consider

$$S = aU + bV$$

$$T = cU + dV$$

$$U = eX + fY$$

$$V = gX + hY$$

Now we determine S and T in terms of X and Y .

We get

$$S = (ae + bg)X + (af + bh)Y$$

$$T = (ce + dg)X + (cf + dh)Y$$

Here we might have noticed that

if we write the equations in matrix form:

$$\begin{pmatrix} S \\ T \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix}$$

$$\begin{pmatrix} U \\ V \end{pmatrix} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

so that we can get

$$\begin{pmatrix} S \\ T \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

Hence we define the matrix multiplication as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$$

2 Why Determinant is defined as that form?

Consider a unit square with coordinates

$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

which places the coordinates of the four vertices anticlockwisely.

Now we perform a transformation A on it

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

The result is

$$\begin{pmatrix} 0 & a & a+b & b \\ 0 & c & c+d & d \end{pmatrix}$$

which also places the coordinates of the four vertices anticlockwisely.

We can find the area of the new shape is

$$ad - bc$$

which is then called the determinant of matrix A .

However, if a matrix has negative determinant, then the way that it places the coordinates of the four vertices will be clockwise.

For example,

$$B = \begin{pmatrix} b & d \\ a & c \end{pmatrix}$$

has determinant $-(ad - bc)$.

If the determinant of some transformation is 0, then the 'dimension' of the original geometric object is 'reduced'.

Actually, the determinant is zero (singular) iff the rows/columns are linearly dependent because

$$\begin{aligned} \det(A) &= ad - bc = 0 \\ \Rightarrow ad &= bc \\ \Rightarrow \frac{a}{b} &= \frac{c}{d} \quad \text{here we just assume } a, b, c, d \text{ are not zero} \end{aligned}$$

3 Inverse

3.1 Claim1

The left inverse L and right inverse R of a matrix A are equal.

Proof:

$$\begin{aligned}
(LA)R &= IR = R \\
&= L(AR) \quad \text{by associativity of matrix multiplication} \\
&= LI = L \\
\Rightarrow L &= R
\end{aligned}$$

3.2 Claim2

A square matrix A has a left inverse $\Leftrightarrow$ it has a right inverse.

Proof:

$$\begin{aligned}
(\Rightarrow) \quad LA &= I \\
\det(LA) &= \det(L) \det(A) = \det(I) = 1 \\
\Rightarrow \det(A) &\neq 0 \Rightarrow A \text{ is invertible} \Rightarrow A^{-1} \text{ exists.} \\
\Rightarrow (LA)A^{-1} &= IA^{-1} \\
\text{Also, } (LA)A^{-1} &= L(AA^{-1}) = LI \\
\Rightarrow IA^{-1} &= LI \Rightarrow L = A^{-1} \\
(\Leftarrow) \quad AR &= I \\
\det(AR) &= \det(A) \det(R) = \det(I) = 1 \\
\Rightarrow \det(A) &\neq 0 \Rightarrow A \text{ is invertible} \Rightarrow A^{-1} \text{ exists.} \\
\Rightarrow A^{-1}(AR) &= A^{-1}I \\
\text{Also, } A^{-1}(AR) &= (A^{-1}A)R = IR \\
\Rightarrow A^{-1}I &= IR \Rightarrow R = A^{-1}
\end{aligned}$$

4 Why the inverse of a 2*2 matrix is defined as that form?

Given a 2*2 matrix M , we can determine its inverse M^{-1} by using $MM^{-1} = I$. After that, we can find it corresponds to the defined form.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\det(M)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Reversing a transformation is exactly convert basis from the original to the priginal vectors before being transformed.